

Year 7 · Chapter 4 · Fractions and percentages: answers (teacher copy)

4.01 Fractions (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: $\frac{7}{4}$
- Ex 1b: $1\frac{3}{4}$
- Ex 2a: $\frac{5}{15}$
- Ex 2b: $\frac{12}{16}$
- Ex 3a: $\frac{5}{6}$
- Ex 3b: $\frac{3}{2} = 1\frac{1}{2}$

Easy (drill)

- | | | |
|---------------------|---------------------|---------------------|
| 1. $\frac{1}{2}$ | 12. $\frac{9}{10}$ | 23. $\frac{57}{10}$ |
| 2. $\frac{11}{12}$ | 13. $\frac{2}{3}$ | 24. $4\frac{5}{12}$ |
| 3. $\frac{1}{3}$ | 14. $\frac{3}{5}$ | 25. $\frac{17}{4}$ |
| 4. $\frac{1}{3}$ | 15. $\frac{1}{10}$ | 26. $\frac{5}{3}$ |
| 5. $\frac{1}{2}$ | 16. $\frac{5}{6}$ | 27. $\frac{19}{10}$ |
| 6. $\frac{1}{2}$ | 17. $\frac{17}{3}$ | 28. $6\frac{1}{3}$ |
| 7. $\frac{1}{12}$ | 18. $\frac{21}{10}$ | 29. $\frac{18}{5}$ |
| 8. $\frac{3}{10}$ | 19. $3\frac{1}{2}$ | 30. $3\frac{2}{5}$ |
| 9. $\frac{5}{8}$ | 20. $\frac{37}{6}$ | 31. $5\frac{1}{2}$ |
| 10. $\frac{1}{3}$ | 21. $\frac{32}{5}$ | 32. $1\frac{1}{2}$ |
| 11. $\frac{11}{12}$ | 22. $2\frac{2}{3}$ | |

Medium (drill)

- | | | |
|-------|-------------------|--------------------|
| 1. 48 | 11. 24 | 19. $\frac{8}{9}$ |
| 2. 28 | 12. 60 | 20. $\frac{2}{3}$ |
| 3. 20 | 13. 5 | 21. $\frac{3}{10}$ |
| 4. 9 | 14. 3 | 22. $\frac{1}{15}$ |
| 5. 4 | 15. 24 | 23. $\frac{3}{5}$ |
| 6. 40 | 16. 6 | 24. $\frac{2}{9}$ |
| 7. 12 | 17. $\frac{1}{2}$ | |
| 8. 20 | 18. $\frac{2}{3}$ | |
| 9. 2 | | |
| 10. 8 | | |

Challenging

- $\frac{5}{6}$
- $\frac{3}{4}$
- $\frac{3}{4}$

Extension

- $\frac{15}{30}$
- $\frac{10}{35}$

Exit ticket

- Level 1: $\frac{17}{5}$
- Level 2: 10
- Level 3: $\frac{3}{4}$

4.02 Ordering fractions (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: $\frac{5}{8}$
- Ex 1b: $\frac{2}{3}$
- Ex 2a: $\frac{3}{10}, \frac{2}{5}, \frac{7}{10}, \frac{4}{5}$
- Ex 2b: $\frac{1}{3}, \frac{1}{2}, \frac{7}{12}, \frac{11}{12}$
- Ex 3a: $\frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$
- Ex 3b: $\frac{5}{4}, 1\frac{1}{3}, \frac{11}{8}$

Easy (drill)

- | | | |
|--------------------|---|---|
| 1. $\frac{3}{4}$ | 11. $\frac{4}{5}$ | 21. $\frac{5}{30}$ and $\frac{12}{30}$ |
| 2. $\frac{7}{10}$ | 12. $\frac{3}{7}$ | 22. $\frac{9}{12}$ and $\frac{1}{12}$ |
| 3. $\frac{1}{5}$ | 13. $\frac{9}{10}$ | 23. $\frac{3}{6}$ and $\frac{4}{6}$ |
| 4. $\frac{6}{10}$ | 14. $\frac{2}{4}$ | 24. $\frac{30}{60}$ and $\frac{54}{60}$ |
| 5. $\frac{5}{6}$ | 15. $\frac{1}{3}$ | 25. $\frac{15}{40}$ and $\frac{36}{40}$ |
| 6. $\frac{9}{10}$ | 16. $\frac{2}{7}$ | 26. $\frac{55}{60}$ and $\frac{48}{60}$ |
| 7. $\frac{11}{12}$ | 17. $\frac{4}{8}$ and $\frac{7}{8}$ | 27. $\frac{6}{10}$ and $\frac{5}{10}$ |
| 8. $\frac{4}{9}$ | 18. $\frac{8}{12}$ and $\frac{6}{12}$ | 28. $\frac{2}{6}$ and $\frac{1}{6}$ |
| 9. $\frac{4}{10}$ | 19. $\frac{21}{30}$ and $\frac{15}{30}$ | |
| 10. $\frac{1}{2}$ | 20. $\frac{9}{15}$ and $\frac{5}{15}$ | |

Medium (drill)

- | | | |
|-------------------|---------------------|--|
| 1. $\frac{9}{12}$ | 8. $\frac{4}{5}$ | 15. $\frac{2}{5}$ |
| 2. $\frac{7}{8}$ | 9. $\frac{7}{10}$ | 16. $\frac{5}{6}$ |
| 3. $\frac{3}{4}$ | 10. $\frac{9}{10}$ | 17. $\frac{2}{4}, \frac{8}{12}, \frac{3}{4}$ |
| 4. $\frac{2}{3}$ | 11. $\frac{10}{12}$ | 18. $\frac{1}{4}, \frac{4}{12}, \frac{10}{12}$ |
| 5. $\frac{1}{2}$ | 12. $\frac{1}{2}$ | 19. $\frac{1}{8}, \frac{2}{5}, \frac{3}{6}$ |
| 6. $\frac{8}{10}$ | 13. $\frac{5}{12}$ | 20. $\frac{1}{3}, \frac{4}{10}, \frac{1}{2}$ |
| 7. $\frac{3}{4}$ | 14. $\frac{4}{5}$ | |

Challenging

- $\frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$
- $\frac{4}{15}, \frac{1}{2}, \frac{3}{5}, \frac{7}{10}$
- $1\frac{1}{6}, \frac{5}{4}, 1\frac{1}{3}, \frac{11}{8}$

Extension

- $\frac{3}{7}, \frac{4}{9}, \frac{1}{2}$
- $\frac{7}{24}$

Exit ticket

- Level 1: $\frac{7}{9}$
- Level 2: $\frac{4}{5}$
- Level 3: $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$

4.03 Adding and subtracting fractions

(booklet p. 11)

WE DO examples (teacher)

- Ex 1a: 1

2. Ex 1b: $\frac{3}{7}$
 3. Ex 2a: $\frac{7}{10}$
 4. Ex 2b: $\frac{11}{20}$
 5. Ex 3a: $\frac{5}{6}$
 6. Ex 3b: $1\frac{7}{12}$

Easy (drill)

- | | | |
|--------------------|---------------------|---------------------|
| 1. 1 | 12. $\frac{1}{4}$ | 22. $\frac{7}{10}$ |
| 2. $1\frac{1}{3}$ | 13. $\frac{5}{12}$ | 23. $1\frac{1}{15}$ |
| 3. 1 | 14. $1\frac{7}{10}$ | 24. $\frac{8}{15}$ |
| 4. 1 | 15. $1\frac{4}{7}$ | 25. $\frac{7}{12}$ |
| 5. $\frac{1}{2}$ | 16. $\frac{3}{4}$ | 26. $1\frac{1}{4}$ |
| 6. $1\frac{2}{5}$ | 17. $1\frac{1}{3}$ | 27. $\frac{4}{5}$ |
| 7. $\frac{8}{11}$ | 18. $\frac{2}{3}$ | 28. 1 |
| 8. $\frac{2}{5}$ | 19. 1 | 29. $\frac{1}{8}$ |
| 9. $1\frac{4}{7}$ | 20. $\frac{1}{4}$ | 30. 1 |
| 10. $\frac{2}{11}$ | 21. $\frac{5}{6}$ | 31. 1 |
| 11. $\frac{1}{12}$ | | 32. $\frac{9}{10}$ |

Medium (drill)

- | | | |
|--------------------|----------------------|---------------------|
| 1. $1\frac{1}{30}$ | 9. $1\frac{1}{6}$ | 17. $\frac{17}{24}$ |
| 2. $\frac{5}{6}$ | 10. $1\frac{13}{30}$ | 18. $\frac{1}{3}$ |
| 3. $\frac{5}{6}$ | 11. $\frac{3}{10}$ | 19. $\frac{1}{2}$ |
| 4. $\frac{3}{8}$ | 12. $\frac{7}{12}$ | 20. $\frac{3}{4}$ |
| 5. $1\frac{1}{12}$ | 13. $\frac{7}{12}$ | 21. $\frac{1}{2}$ |
| 6. $\frac{1}{4}$ | 14. $\frac{5}{8}$ | 22. $\frac{2}{3}$ |
| 7. $1\frac{2}{5}$ | 15. $\frac{21}{40}$ | |
| 8. $1\frac{7}{15}$ | 16. $\frac{1}{10}$ | |

Challenging

1. $1\frac{19}{30}$
 2. $1\frac{1}{12}$
 3. $\frac{3}{8}$

Extension

1. $\frac{11}{15}$
 2. $\frac{1}{3} + \frac{1}{4}$

Exit ticket

1. Level 1: $\frac{5}{7}$
 2. Level 2: $\frac{1}{3}$
 3. Level 3: $1\frac{4}{15}$

4.04 Adding and subtracting mixed numerals (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: $8\frac{7}{10}$
 2. Ex 1b: $5\frac{1}{4}$
 3. Ex 2a: $9\frac{2}{3}$

4. Ex 2b: $1\frac{1}{2}$
 5. Ex 3a: $1\frac{7}{12}$
 6. Ex 3b: $2\frac{13}{20}$

Easy (drill)

- | | | |
|----------------------|--------------------|---------------------|
| 1. $2\frac{1}{2}$ | 12. $6\frac{1}{2}$ | 23. $1\frac{1}{10}$ |
| 2. 13 | 13. 5 | 24. $4\frac{1}{3}$ |
| 3. 9 | 14. $2\frac{1}{3}$ | 25. $6\frac{1}{7}$ |
| 4. $5\frac{3}{4}$ | 15. 14 | 26. $6\frac{2}{5}$ |
| 5. $5\frac{1}{5}$ | 16. 3 | 27. $6\frac{1}{3}$ |
| 6. 2 | 17. $7\frac{8}{9}$ | 28. $6\frac{3}{4}$ |
| 7. $6\frac{1}{5}$ | 18. $4\frac{1}{6}$ | 29. $2\frac{1}{3}$ |
| 8. $6\frac{1}{3}$ | 19. $3\frac{1}{2}$ | 30. $6\frac{4}{9}$ |
| 9. $12\frac{2}{3}$ | 20. $8\frac{3}{4}$ | 31. $1\frac{1}{2}$ |
| 10. 5 | 21. $4\frac{1}{5}$ | 32. $2\frac{4}{5}$ |
| 11. $11\frac{1}{10}$ | 22. $7\frac{3}{8}$ | |

Medium (drill)

- | | | |
|---------------------|----------------------|--------------------|
| 1. $5\frac{9}{10}$ | 7. $7\frac{11}{24}$ | 13. $1\frac{1}{2}$ |
| 2. $11\frac{1}{4}$ | 8. $6\frac{14}{15}$ | 14. $6\frac{4}{5}$ |
| 3. $8\frac{7}{10}$ | 9. $8\frac{11}{12}$ | 15. $2\frac{1}{2}$ |
| 4. $4\frac{5}{8}$ | 10. $7\frac{23}{24}$ | 16. $4\frac{2}{5}$ |
| 5. $7\frac{3}{4}$ | 11. $7\frac{5}{8}$ | 17. $4\frac{2}{3}$ |
| 6. $7\frac{17}{30}$ | 12. $8\frac{1}{4}$ | 18. $3\frac{2}{3}$ |

Challenging

1. $1\frac{7}{12}$
 2. $1\frac{7}{10}$
 3. $6\frac{7}{12}$

Extension

1. $1\frac{1}{12}$
 2. $3\frac{1}{2}$

Exit ticket

1. Level 1: $6\frac{2}{3}$
 2. Level 2: $5\frac{3}{5}$
 3. Level 3: $1\frac{7}{12}$

4.05 Fraction of a quantity (booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: 7
 2. Ex 1b: 9
 3. Ex 2a: 16 h
 4. Ex 2b: 1600 m
 5. Ex 3a: 135 min
 6. Ex 3b: 50 s

Easy (drill)

- | | | |
|-------|------|--------|
| 1. 6 | 5. 2 | 9. 3 |
| 2. 3 | 6. 3 | 10. 7 |
| 3. 6 | 7. 3 | 11. 8 |
| 4. 12 | 8. 8 | 12. 11 |

- | | | |
|--------|--------|--------|
| 13. 9 | 20. 15 | 27. 33 |
| 14. 9 | 21. 21 | 28. 18 |
| 15. 2 | 22. 16 | 29. 16 |
| 16. 4 | 23. 36 | 30. 44 |
| 17. 50 | 24. 32 | 31. 40 |
| 18. 14 | 25. 30 | 32. 21 |
| 19. 72 | 26. 70 | |

Medium (drill)

- | | | |
|-----------|--------------------|---------------------|
| 1. 760 g | 9. 16 h | 16. $\frac{2}{3}$ |
| 2. 500 g | 10. 12 h | 17. $\frac{1}{6}$ |
| 3. 18 h | 11. 250 m | 18. $\frac{1}{5}$ |
| 4. 50 g | 12. 250 g | 19. $\frac{8}{15}$ |
| 5. 36 min | 13. $\frac{4}{5}$ | 20. $\frac{13}{24}$ |
| 6. 20 cm | 14. $\frac{1}{5}$ | |
| 7. 375 m | 15. $\frac{5}{12}$ | |
| 8. 30 min | | |

Challenging

- 12 boys
- \$240
- 36

Extension

- 33.75
- 160 L

Exit ticket

- Level 1: 9
- Level 2: 60
- Level 3: 90 min

4.06 Multiplying fractions (booklet p. 23)

WE DO examples (teacher)

- Ex 1a: $\frac{2}{15}$
- Ex 1b: 6
- Ex 2a: $\frac{5}{14}$
- Ex 2b: $\frac{1}{40}$
- Ex 3a: $\frac{3}{4}$
- Ex 3b: $10\frac{8}{9}$

Easy (drill)

- | | | |
|--------------------|--------------------|--------------------|
| 1. $\frac{5}{36}$ | 11. $\frac{2}{5}$ | 22. $7\frac{6}{7}$ |
| 2. $\frac{1}{9}$ | 12. $\frac{3}{10}$ | 23. 6 |
| 3. $\frac{6}{35}$ | 13. $\frac{2}{5}$ | 24. $3\frac{1}{3}$ |
| 4. $\frac{7}{16}$ | 14. $\frac{5}{36}$ | 25. 1 |
| 5. $\frac{4}{25}$ | 15. $\frac{2}{9}$ | 26. $5\frac{1}{2}$ |
| 6. $\frac{1}{18}$ | 16. $\frac{1}{24}$ | 27. $2\frac{3}{4}$ |
| 7. $\frac{1}{4}$ | 17. 1 | 28. $1\frac{1}{5}$ |
| 8. $\frac{5}{56}$ | 18. 2 | 29. $2\frac{1}{2}$ |
| 9. $\frac{1}{4}$ | 19. $2\frac{2}{5}$ | 30. $1\frac{3}{7}$ |
| 10. $\frac{1}{10}$ | 20. $1\frac{3}{5}$ | 31. 4 |
| | 21. $7\frac{1}{5}$ | 32. $3\frac{1}{9}$ |

Medium (drill)

- | | | |
|-------------------|--------------------|-------------------|
| 1. $\frac{3}{20}$ | 3. $\frac{3}{14}$ | 5. $\frac{2}{21}$ |
| 2. $\frac{5}{28}$ | 4. $\frac{16}{35}$ | 6. $\frac{3}{5}$ |

- | | | |
|--------------------|--------------------|---------------------|
| 7. $\frac{1}{15}$ | 12. $\frac{1}{6}$ | 17. $12\frac{5}{6}$ |
| 8. $\frac{1}{6}$ | 13. $\frac{1}{3}$ | 18. $5\frac{1}{15}$ |
| 9. $\frac{5}{28}$ | 14. $\frac{1}{5}$ | 19. $3\frac{8}{9}$ |
| 10. $\frac{1}{14}$ | 15. $\frac{4}{35}$ | 20. $3\frac{9}{10}$ |
| 11. $\frac{4}{49}$ | 16. $\frac{2}{15}$ | 21. 4 |
| | | 22. $2\frac{7}{9}$ |

Challenging

- $10\frac{8}{9}$
- $2\frac{1}{3}$
- $1\frac{1}{6}$

Extension

- $\frac{1}{10}$
- $9\frac{1}{3} \text{ m}^2$

Exit ticket

- Level 1: $\frac{3}{20}$
- Level 2: $\frac{1}{6}$
- Level 3: $3\frac{1}{3}$

4.07 Dividing fractions (booklet p. 27)

WE DO examples (teacher)

- Ex 1a: $\frac{4}{3}$, $\frac{1}{5}$, $\frac{2}{5}$
- Ex 1b: 50
- Ex 2a: $\frac{1}{6}$
- Ex 2b: 2
- Ex 3a: 2
- Ex 3b: $1\frac{1}{2}$

Easy (drill)

- | | | |
|-------------------|--------------------|--------|
| 1. $\frac{2}{10}$ | 10. $\frac{8}{7}$ | 19. 12 |
| 2. $\frac{1}{5}$ | 11. $\frac{1}{3}$ | 20. 36 |
| 3. $\frac{3}{4}$ | 12. $\frac{1}{6}$ | 21. 40 |
| 4. $\frac{10}{5}$ | 13. $\frac{7}{5}$ | 22. 15 |
| 5. $\frac{7}{4}$ | 14. $\frac{2}{8}$ | 23. 42 |
| 6. $\frac{1}{4}$ | 15. $\frac{6}{4}$ | 24. 49 |
| 7. $\frac{1}{8}$ | 16. $\frac{12}{3}$ | 25. 20 |
| 8. $\frac{4}{10}$ | 17. 12 | 26. 12 |
| 9. 6 | 18. 14 | 27. 25 |
| | | 28. 18 |
| | | 29. 24 |
| | | 30. 16 |
| | | 31. 21 |
| | | 32. 48 |

Medium (drill)

- | | | |
|-------------------|---------------------|----------------------|
| 1. 2 | 9. $2\frac{1}{2}$ | 16. $\frac{9}{14}$ |
| 2. $\frac{2}{3}$ | 10. $2\frac{2}{3}$ | 17. $1\frac{2}{5}$ |
| 3. $\frac{7}{18}$ | 11. $1\frac{1}{8}$ | 18. $1\frac{11}{14}$ |
| 4. $\frac{1}{2}$ | 12. $2\frac{1}{4}$ | 19. 1 |
| 5. $2\frac{1}{2}$ | 13. $\frac{1}{2}$ | 20. $1\frac{1}{2}$ |
| 6. $3\frac{1}{3}$ | 14. $\frac{5}{21}$ | 21. $1\frac{1}{25}$ |
| 7. $\frac{1}{2}$ | 15. $\frac{20}{21}$ | 22. $2\frac{2}{15}$ |
| 8. 1 | | |

Challenging

- 3
- 2
- $1\frac{1}{2}$

Extension

- $\frac{2}{3}$
- 5

Exit ticket

- Level 1: $\frac{8}{5}$
- Level 2: 4
- Level 3: 2

4.08 Fraction problems (booklet p. 31)

WE DO examples (teacher)

- Ex 1a: $6\frac{7}{12}$ L
- Ex 1b: $1\frac{1}{4}$ L
- Ex 2a: $\frac{1}{12}$
- Ex 2b: 1 goat
- Ex 3a: $\frac{4}{15}$
- Ex 3b: 64 pages

Easy (drill)

- | | | |
|------------------|------------------|------------------|
| 1. $\frac{2}{5}$ | 6. $\frac{2}{3}$ | 13. 18 kilograms |
| 2. $\frac{1}{4}$ | 7. $\frac{1}{3}$ | 14. 10 minutes |
| 3. $\frac{1}{3}$ | 8. $\frac{2}{5}$ | 15. \$10 |
| 4. $\frac{1}{6}$ | 9. $\frac{3}{5}$ | 16. 8 minutes |
| 5. $\frac{4}{5}$ | 10. 6 kilograms | 17. 9 kilograms |
| | 11. \$3 | 18. 21 minutes |
| | 12. \$4 | 19. \$14 |
| | | 20. \$18 |
| | | 21. \$50 |

Medium (drill)

- | | | |
|---------|----------|--------|
| 1. \$6 | 6. \$4 | 11. 18 |
| 2. \$15 | 7. \$27 | 12. 70 |
| 3. \$16 | 8. \$12 | 13. 6 |
| 4. \$57 | 9. \$40 | 14. 10 |
| 5. \$33 | 10. \$28 | |

Challenging

- 100
- $4\frac{9}{10}$ m
- 63

Extension

- 24
- 300 mL

Exit ticket

- Level 1: $\frac{11}{15}$
- Level 2: 9
- Level 3: \$70

4.09 Fractions on a calculator (booklet p. 37)

WE DO examples (teacher)

- Ex 1a: $\frac{11}{12}$
- Ex 1b: $1\frac{2}{5}$
- Ex 2a: $3\frac{2}{5}$; 3.4

- Ex 2b: $\frac{25}{72}$
- Ex 3a: $12\frac{1}{4}$
- Ex 3b: $5\frac{1}{16}$

Easy (drill)

- | | | |
|----------------------|-----------------------|-----------------------|
| 1. $\frac{3}{22}$ | 12. $\frac{37}{171}$ | 23. $1\frac{1}{8}$ |
| 2. $\frac{17}{24}$ | 13. $\frac{19}{52}$ | 24. $\frac{2}{3}$ |
| 3. $\frac{5}{8}$ | 14. 1 | 25. $1\frac{81}{280}$ |
| 4. $1\frac{71}{234}$ | 15. $1\frac{1}{14}$ | 26. $\frac{54}{187}$ |
| 5. $\frac{35}{247}$ | 16. $\frac{13}{45}$ | 27. $\frac{7}{39}$ |
| 6. $\frac{23}{40}$ | 17. $\frac{4}{35}$ | 28. $\frac{2}{75}$ |
| 7. $\frac{13}{15}$ | 18. $\frac{121}{126}$ | 29. $\frac{6}{35}$ |
| 8. $\frac{53}{84}$ | 19. $\frac{8}{35}$ | 30. $\frac{12}{91}$ |
| 9. $\frac{181}{240}$ | 20. $\frac{270}{323}$ | 31. $\frac{3}{16}$ |
| 10. $\frac{77}{190}$ | 21. $1\frac{5}{22}$ | 32. $\frac{16}{45}$ |
| 11. $1\frac{11}{21}$ | 22. $\frac{9}{112}$ | |

Medium (drill)

- | | | |
|---------------------|----------------------|----------------------|
| 1. $\frac{2}{3}$ | 8. $\frac{1}{3}$ | 15. $1\frac{5}{13}$ |
| 2. $\frac{5}{12}$ | 9. $\frac{5}{8}$ | 16. $6\frac{7}{12}$ |
| 3. $1\frac{13}{72}$ | 10. $\frac{37}{144}$ | 17. $1\frac{13}{15}$ |
| 4. $\frac{55}{98}$ | 11. $\frac{23}{56}$ | 18. $8\frac{4}{15}$ |
| 5. $\frac{11}{49}$ | 12. $\frac{49}{54}$ | 19. $4\frac{5}{9}$ |
| 6. $\frac{57}{280}$ | 13. $8\frac{8}{11}$ | 20. $2\frac{8}{11}$ |
| 7. $\frac{8}{21}$ | 14. $3\frac{2}{3}$ | |

Challenging

- $12\frac{1}{4}$
- $5\frac{9}{16}$
- $2\frac{4}{15}$

Extension

- $\frac{31}{32}$: it gets closer to 1
- $\frac{13}{15}$

Exit ticket

- Level 1: $\frac{31}{35}$
- Level 2: $3\frac{1}{6}$
- Level 3: 4

4.10 Percentages, fractions and decimals

(booklet p. 41)

WE DO examples (teacher)

- Ex 1a: $\frac{13}{100}$; 0.13
- Ex 1b: $\frac{2}{25}$; 0.08
- Ex 2a: $\frac{3}{40}$; 0.075
- Ex 2b: 70%
- Ex 3a: 92%
- Ex 3b: 4.5%

Easy (drill)

- | | | |
|--------------------|--------------------|----------|
| 1. $\frac{1}{4}$ | 10. $\frac{3}{25}$ | 20. 0.22 |
| 2. $\frac{3}{4}$ | 11. $\frac{9}{10}$ | 21. 0.62 |
| 3. $\frac{7}{20}$ | 12. $\frac{3}{5}$ | 22. 0.68 |
| 4. $\frac{2}{25}$ | 13. $\frac{1}{5}$ | 23. 1.1 |
| 5. $\frac{9}{20}$ | 14. $\frac{1}{10}$ | 24. 0.1 |
| 6. $\frac{1}{100}$ | 15. $\frac{1}{25}$ | 25. 0.14 |
| 7. $\frac{4}{5}$ | 16. $\frac{7}{10}$ | 26. 0.47 |
| 8. $\frac{3}{20}$ | 17. 0.99 | 27. 0.35 |
| 9. $\frac{2}{5}$ | 18. 1.11 | 28. 0.59 |
| | 19. 0.34 | 29. 0.76 |
| | | 30. 1.17 |
| | | 31. 0.46 |
| | | 32. 1.18 |

Medium (drill)

- | | | |
|--------|-------------------------|-------------------------|
| 1. 60% | 10. 40% | 18. 40%, $\frac{2}{5}$ |
| 2. 32% | 11. 65% | 19. 20%, $\frac{1}{5}$ |
| 3. 70% | 12. 92% | 20. 12%, $\frac{3}{25}$ |
| 4. 34% | 13. 16% | 21. 50%, $\frac{1}{2}$ |
| 5. 80% | 14. 26% | 22. 75%, $\frac{3}{4}$ |
| 6. 36% | 15. 76% | |
| 7. 50% | 16. 42% | |
| 8. 10% | 17. 30%, $\frac{3}{10}$ | |
| 9. 25% | | |

Challenging

- 0.375, 37.5%
- $33\frac{1}{3}\%$
- $\frac{1}{8}$

Extension

- $66\frac{2}{3}\%$
- $\frac{37}{300}$

Exit ticket

- Level 1: $\frac{1}{4}$
- Level 2: 7%
- Level 3: 18.75%

4.11 Percentage of a quantity (booklet p. 45)**WE DO examples (teacher)**

- Ex 1a: 35
- Ex 1b: 21
- Ex 2a: \$9
- Ex 2b: 84
- Ex 3a: 60 kg
- Ex 3b: \$175

Easy (drill)

- | | | |
|--------|---------|---------|
| 1. 28 | 12. 6 | 23. 29 |
| 2. 52 | 13. 26 | 24. 26 |
| 3. 84 | 14. 4 | 25. 45 |
| 4. 17 | 15. 59 | 26. 156 |
| 5. 25 | 16. 46 | 27. 180 |
| 6. 11 | 17. 24 | 28. 17 |
| 7. 29 | 18. 72 | 29. 64 |
| 8. 14 | 19. 78 | 30. 8 |
| 9. 2 | 20. 20 | 31. 435 |
| 10. 36 | 21. 112 | 32. 13 |
| 11. 14 | 22. 14 | |

Medium (drill)

- | | | |
|------------|------------|------------|
| 1. \$72 | 9. \$49.5 | 17. \$97.5 |
| 2. \$36 | 10. \$23 | 18. \$198 |
| 3. \$212.5 | 11. \$82.5 | 19. \$234 |
| 4. \$66 | 12. \$35 | 20. \$54 |
| 5. \$186 | 13. \$15 | 21. \$20 |
| 6. \$88 | 14. \$259 | 22. \$112 |
| 7. \$95 | 15. \$315 | |
| 8. \$72.5 | 16. \$52.5 | |

Challenging

- \$175
- 9
- \$46

Extension

- \$57.60
- No: 20% then 10% is 28% off

Exit ticket

- Level 1: 8
- Level 2: 70
- Level 3: \$51

4.12 Expressing as a fraction or percentage (booklet p. 49)**WE DO examples (teacher)**

- Ex 1a: $\frac{17}{20}$; 85%
- Ex 1b: 75%
- Ex 2a: $\frac{1}{5}$
- Ex 2b: 20%
- Ex 3a: 37.5%
- Ex 3b: Mia 85%, Jake 84%: Mia

Easy (drill)

- | | | |
|---------|---------------------|---------------------|
| 1. 10% | 15. 55% | 24. $\frac{11}{20}$ |
| 2. 50% | 16. 56% | 25. $\frac{13}{24}$ |
| 3. 75% | 17. $\frac{3}{10}$ | 26. $\frac{11}{20}$ |
| 4. 50% | 18. $\frac{5}{16}$ | 27. $\frac{11}{15}$ |
| 5. 25% | 19. $\frac{1}{2}$ | 28. $\frac{11}{12}$ |
| 6. 20% | 20. $\frac{3}{20}$ | 29. $\frac{1}{16}$ |
| 7. 40% | 21. $\frac{17}{30}$ | 30. $\frac{13}{16}$ |
| 8. 90% | 22. $\frac{11}{24}$ | 31. $\frac{9}{20}$ |
| 9. 92% | 23. $\frac{29}{30}$ | 32. $\frac{1}{3}$ |
| 10. 36% | | |
| 11. 60% | | |
| 12. 32% | | |
| 13. 20% | | |
| 14. 30% | | |

Medium (drill)

- | | | |
|----------|-----------|-----------|
| 1. 5% | 8. 50% | 15. 75% |
| 2. 25% | 9. 10% | 16. 42.5% |
| 3. 10% | 10. 37.5% | 17. 97.5% |
| 4. 3.33% | 11. 6.25% | 18. 35% |
| 5. 2.5% | 12. 25% | 19. 62% |
| 6. 15% | 13. 80% | 20. 85% |
| 7. 20% | 14. 80% | |

Challenging

- Mia (85% vs 84%)
- 25%
- 25%

Extension

- 24%
- 20%

Exit ticket

1. Level 1: $\frac{3}{4}$

2. Level 2: 84%

3. Level 3: 20%